

Final Project: Customer Churn **Prediction** **(Use Case Study with Deployment** **and Presentation)**

Course: B.Tech – Gen AI (2nd Semester)

Project Type: Individual Final Project

Dataset: customer_churn.csv (attached in Classroom)

Required Deliverables:

- Jupyter Notebook (.ipynb)
 - Streamlit Deployment Link
 - Final Project Presentation (PPT)
 - Dataset File Used
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Project Title

Customer Churn Prediction using Random Forest: End-to-End Machine Learning and Deployment Use Case

Project Description

This project aims to provide hands-on experience in developing a complete Machine Learning solution, starting from data preprocessing to model development and online deployment. Students will work with a real-world customer churn dataset and build a classification model using the Random Forest algorithm. The project includes exploratory data analysis, visualization, model evaluation, deployment using Streamlit Cloud, and preparation of a formal presentation summarizing the entire workflow.

This assignment simulates an industry-style project where students are expected to produce a deployable AI solution supported by analytical insights.

Learning Objectives

Upon successful completion of this project, students will be able to:

1. Understand and apply data preprocessing techniques.
 2. Perform feature engineering and transformation on structured data.
 3. Develop, train, and evaluate a classification model using Random Forest.
 4. Generate meaningful visualizations using matplotlib and related libraries.
 5. Deploy a machine learning model using Streamlit Cloud.
 6. Present project outcomes effectively through a structured PPT.
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Dataset Description

The dataset provided (customer_churn.csv) includes information related to:

- Customer demographics
- Account details
- Service usage statistics
- Churn status (Yes/No)

Students must load and utilize this dataset in their analysis and model-building process.

Project Requirements

(Mandatory Sections in Notebook)

1. Introduction

The introduction section must explain:

- The concept of customer churn
- The relevance of churn prediction for businesses
- Why Random Forest is used in this project

2. Importing Required Libraries

Students must import all essential libraries, including but not limited to:

- pandas, numpy
- scikit-learn
- matplotlib, seaborn
- scikit-plot (optional)

3. Data Loading and Exploration

- Load the dataset
- Display initial rows
- Provide summary statistics
- Check for missing values and data types

4. Data Cleaning and Preprocessing

- Handle missing or inconsistent data
- Encode categorical features
- Convert target labels (Yes/No) to binary values
- Split data into training and testing sets

5. Model Development: Random Forest Classifier

- Train the RandomForestClassifier
- Generate predictions using the test dataset

6. Model Evaluation

Model performance must be evaluated using:

- Accuracy score
- Confusion matrix
- Classification report
- ROC curve
- Feature importance analysis

7. Visualizations

At minimum, the notebook should include:

- Churn distribution visualization
- Visualizations of key features
- ROC curve
- Feature importance bar chart

8. Conclusion

Students must summarize:

- Model performance and interpretation
 - Features contributing significantly to churn
 - Possible improvements or future enhancements
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Model Deployment Using Streamlit Cloud **(Mandatory)**

Each student must deploy their trained model as a functional web application.

Deployment Requirements

Files required:

1. app.py
2. model.pkl
3. requirements.txt

The requirements.txt file must include:

```
streamlit  
pandas  
numpy  
scikit-learn  
scikit-plot  
matplotlib
```

Streamlit Deployment Procedure

Step 1: Prepare a GitHub Repository

Upload the following files into a new GitHub repository:

- app.py
- model.pkl
- requirements.txt

Step 2: Access Streamlit Cloud

Visit the Streamlit Cloud platform and sign in with your GitHub account.

Step 3: Deploy the Application

- Click on "Deploy App"
- Select your repository
- Choose `app.py` as the main file
- Confirm and deploy

Streamlit will automatically build and deploy the application.

Step 4: Submit Deployment Link

Upon deployment, Streamlit will generate a public web link in the following format:

`https://your-app-name.streamlit.app/`

This link must be included in your final submission.

Final PPT Presentation Requirement **(Mandatory)**

In addition to the Jupyter Notebook and deployed app, students must prepare a professional PPT summarizing the entire project.

A detailed **PPT structure and guideline document is attached for reference.**

Follow it strictly when preparing your presentation.

Presentation Requirements

Your PPT must include:

1. **Title Slide**
 - Project title
 - Student name and enrollment number
 - Course and instructor details
2. **Project Overview**
 - Introduction to churn prediction
 - Purpose and goals
3. **Dataset Description**
 - Overview of features
 - Size and structure of data
4. **Machine Learning Workflow**
 - Data cleaning
 - Feature encoding
 - Model training
5. **Model Evaluation**
 - Accuracy
 - Confusion matrix
 - ROC curve
 - Feature importance
6. **Visualizations**
 - Churn distribution
 - Feature analysis
7. **Deployment Section**
 - Summary of deployment steps
 - Screenshot of the working Streamlit app
8. **Conclusion**
 - Key insights and findings
 - Suggestions for improvement

Final Submission Checklist

Students must submit the following:

1. Jupyter Notebook (.ipynb)
2. Streamlit Deployment Link
3. Final PPT Presentation (.pptx)
4. Dataset file used in the project

Note: A complete PPT Guide has been attached separately for students to follow while preparing their presentations.